Handoff: Babbage's Hall

For a new chat: how to work, build, test, and what's next · 31 October 2026

Read this first in a new chat. It says how William likes to work, where everything is, how to build and test, what has been learned, and what to do next. The list of machines is `MACHINES.md`; the hall itself is described in `README.md`. This file is updated whenever a chat ends.

1. What this is

- **Babbage's Hall** (`demo/WyomingWill/BabbagesHall/`), a Victorian engineering hall in the Bullet Physics Playground (bpp, **github.com/bullet-physics-playground/bpp**, branch `master`): cast iron, brass, a glass roof, holding **machines that compute**. A separate museum from William's Euler's Gallery (`../EulersGallery/`), built on its machinery (separate worlds, sandboxes, walking with Tab).
- **Every machine can be 3D printed and built.** Each exhibit's parts are OpenSCAD sources (the single source for the simulation and the prints), a `make-stls.sh` makes the STL files, and its README lists what to buy and how to assemble it. What is printed is what is simulated: no shortcuts in the parts that compute, and anything idealised is said on the placard and in FINDINGS.
- **The walk gets more interesting as you go**, least interesting first, Babbage's Difference Engine at the far end (`MACHINES.md` ranks them).
- Built so far: **the Differential** (west 1), **Lehmer's number sieve** (east 1), **the Pascaline** (west 2), **the Domino Half Adder** (east 2), and the **Digi-Comp II** (west 3: it counts, adds, multiplies, subtracts and divides, runs programs, and in AUTO runs a multiply by itself; OVERFLOW's halt is next). Covered bays wait for Kelvin's tide predictor (east 3), the Differential Analyser (west 4) and the Difference Engine (east 4).
- William M. Spears (`williammspears@gmail.com`) owns the WyomingWill demos; Jakob (koppi) wrote bpp. Clocks are William's own separate collection.

2. How William likes to work

- **Deliver every change as a patch plus the git commands.** He applies patches in a checkout called `bpp-push`, pushes, then pulls in his installation checkout (which has `pull.ff only`; never apply patches there). Give the commands as below, and say if bpp must be rebuilt (only when C++ changed).

```
git pull
git am ~/Downloads/<name>.patch
git push
```

- **Every patch includes all its documents, PDFs too.** Every `.md` has a typeset `.pdf` beside it (`./make-pdfs.sh`, or `tools/md2pdf.py` for one); rebuild the PDF whenever its `.md` changes. Euler's Gallery and the Rec Room have README.pdfs made with `../ArtificialPhysics/tools/md2pdf.py`.
- **Before sending a patch:** check it applies to the current `origin/master` (`git fetch` first; a patch built on one he hasn't applied fails, as the first `MACHINES.md` patch did); no trailing spaces (`git diff --check`); `tools/check-all.sh` passes; the visuals looked at (`tools/shots.sh`). A re-sent patch gets a new file name, so a stale download can't be applied.

- Commits are authored `William M. Spears <williammspears@gmail.com>`, with the Co-Authored-By trailer for Claude. Stage only the files you changed.
- He looks at the exhibits in the GUI and reports what he sees with screenshots: take his reports seriously and reproduce them before answering, and measure when the pictures don't settle it.
- Plain explanations; measure before claiming; say what was and wasn't tested.
- **His bpp is built with Lua 5.2;** a cloud workspace's is usually 5.1. Write Lua that runs on both: no bare `setfenv`/`getfenv` (give `loadfile` its environment with `"t", _ENV` on 5.2, as the exhibits' table loaders do), no `goto`. `tools/check-lua52.sh` checks the one place this has bitten (the table loader) under a standalone `lua5.2`.

3. Where everything is

Path	What
`hall.lua`	The hall: architecture, the EXHIBITS list (walking order, placards, `floorY`), sandboxes, walking and camera flights, resting, the cost meter
`exhibit-table.lua`	The standard exhibit table (README "Exhibit standard")
`MACHINES.md`	All 47 machines considered, ranked by interest and scored for complexity, with how each would go in bpp and be printed; the rope test
`README.md`	The hall: bays, keys, how it works, the cost meter, the exhibit standard, notes for exhibit authors
`HANDOFF.md`	This file
`make-pdfs.sh`, `tools/md2pdf.py`	The PDFs
`make-labels.py`, `hall-meshes/`	The placards' lettering (Pillow, NumPy, DejaVu fonts)
`tools/check-all.sh`	Every exhibit's tour, on its own and in the hall
`tools/shots.sh`, `tools/keys.sh`	Screenshots on a virtual screen; with keys, hover, camera flights
`tools/check-lua52.sh`	The Lua 5.2 sandbox check
`tools/rope-test.lua`	The rope experiment (for the tide predictor)
`exhibits/DigiComp2/tests/toggle.lua`, `tests/switch.lua`, `tests/junction.lua`	One Digi-Comp flip-flop alone (with its foot, done hole and clear lane); one switch; the piece of board where the memory meets the accumulator
`exhibits/DigiComp2/MANUAL.md`, `manual/`	The Digi-Comp's illustrated manual; `manual/make-figures.sh` redraws its figures from the geometry (lua, python3, matplotlib). `tools/md2pdf.py` puts a `.md` file's images into its PDF
`exhibits/<Name>/`	One folder per exhibit: its script, `parts/` (OpenSCAD), `make-stls.sh`, `README.md`/`.pdf`, `FINDINGS.md`/`.pdf`

Related: `../EulersGallery/` (the gallery the hall's machinery came from; its tables still follow its own sizes, see §7), `../RecRoom/` (games room: the cost meter and suspending came from there), `demo/claude/digicomp.lua` (an incomplete Digi-Comp II, a starting point), `module/scad/gearsv50.lua` and the Differential's `parts/gearsv50.scad` (bevel gears).

4. Building, running, testing

```
git clone https://github.com/bullet-physics-playground/bpp ~/bpp && cd ~/bpp
apt-get update
```

```
apt-get install -y libbullet-dev libluabind-dev liblua5.1-0-dev liblua5.2-dev \
  libqglviewer-dev-qt5 libsdl2-mixer-dev libassimp-dev qtbase5-dev libglew-dev \
  xvfb xdotool imagemagick openscad lua5.2 povray
pip install --break-system-packages trimesh manifold3d markdown playwright
mkdir -p build && cd build && qmake ../bpp.pro CONFIG+=release && make -j2 release
```

That gives `build/release/bpp` (a few minutes to ten). The tools find it there, in `release/`, or on the path; or set `BPP`.

- **Headless:** from the top folder, `build/release/bpp -f demo/WyomingWill/BabbagesHall/hall.lua -n 600`. `HALL_START=pascaline` starts at an exhibit. The exhibits run on their own the same way, each with a tour that prints `RESULT <name> tour: PASS`.
- **GUI on a virtual screen:** `tools/shots.sh` (see its header); `LIBGL_ALWAYS_SOFTWARE=1`, so drawing is slow and the simulation runs slower than real time there. In the GUI, `print` goes to bpp's console pane, not the terminal: write to a file (the meter's `HALL_METER_FILE`, the flights' `HALL_FLY_LOG=file`) to read it.
- **Printed parts:** each exhibit's `make-stls.sh` (OpenSCAD 2021.01 here; William's is newer: avoid `assign()` and `polyhedron(triangles=)`). Check every STL is one watertight solid (trimesh), and that parts don't overlap at rest (manifold3d); OpenSCAD renders pictures under `xvfb-run`.
- **POV-Ray:** `build/release/bpp -f ... -n 2 -e` writes `export/<script>/` (untracked; delete it after); render with `povray +I<script>.pov +K2 +W800 +H450 +L<bpp folder>`.

5. What has been learned (bpp and Bullet)

- **OpenSCAD parts** are cached by the text asked for, not by the files' contents: put a checksum of every file the part includes into the text (each exhibit's `checksum()`). Set `collides` before `v:add`, the transform after. A body's origin must be its centre of mass.
- **Two bpp processes building the same new part at once** can collide in the mesh cache: run tests one at a time after a part changes.
- **Gears:** mesh-on-mesh (GImpact) bevel teeth jam; a compound of convex hulls (one per tooth plus the body, from the same SCAD source) meshes well (`exhibits/Differential/FINDINGS.md`).
- **Hinge angles** (`getHingeAngle`) are clockwise-positive as seen down the hinge's axis, the opposite of a right-handed turn. A hinge's motor speed counts the same way as its angle (measured on the Digi-Comp's wheel).
- **A pin's friction** can be had from a hinge's motor held at no speed with a small impulse a step (`enableAngularMotor(true, 0, torque * step)`); the Digi-Comp's rockers use it, from PLA on PLA at the boss.
- **Over-centre parts** (a rocker whose centre of mass is above its pin) rest in a very shallow dip: with no friction they bounced back out of it, with pin friction they stop wherever they're left. Give them weight high up (the Digi-Comp's rockers have a steel ball pressed in) and test both extremes.
- **Walls as boxes** run past their ends by half their thickness if made `length + thickness` long, so the joints are solid: check where such an end meets a moving part (the Digi-Comp's feeder wheel rubbed two), or narrows a channel (two jammed the Digi-Comp's clear lane). Mitre a lane's walls at its bends (`laneWalls` in the Digi-Comp's geometry).
- **A ball pushing a part out of its way** pushes it only until it can squeeze past, which may be nowhere near where the part should end up: to push it all the way, guide the ball (the Digi-Comp's clear lane narrows beside each foot).

- **Two ways into one place:** trace the ball that comes in first (its position every frame) before giving the other one a way in; a ball rolling down a wall must pass under the ends of the walls above it with room for the ball (the Digi-Comp's joining funnels, `FINDINGS.md`). Lanes on one level can't cross: their order is fixed by where they start and end, so plan a board in a drawing of every wall first.
- **A part with holes** (the Digi-Comp's upper plate) is a static body of boxes round the holes, drawn not by them (`transparency = 1`) but by the OpenSCAD part. `transparency` near 0.35 draws a see-through plate well.
- **Luabind objects** from `v:eachContact` can't be compared with `==`; look them up as table keys (`names[obj]`).
- **Small steps:** the exhibits take 2400 steps a simulated second, stepped by hand in `preSim` (`common.setTiming(STEP, 0, STEP)`); the teeth and the dominoes need it. Sleeping is off for moving parts (state 4), so each exhibit **rests** (stops stepping) once nothing moves; it changes nothing computed.
- **Springs** that matter are simulated as the printed part: the Pascaline's detent is a body on a hinge turned back by the leaf's stiffness from PLA (3EI/L), and its working range of thickness was found by sweeping it.
- **Rope:** none built in; a rope of beads on ball joints works if the rope isn't far lighter than what hangs on it (`MACHINES.md`, "Rope").
- **Mouse:** scripts get `onHover` (after half a second still, no button down) but no clicks; keys override bpp's own one-letter keys.
- `btTransform` has no `getOrigin` in bpp's Lua; `.trans`, `:mult` and `getCenterOfMassTransform` are what there is.
- **Compounds:** a `btCompoundShape` in bpp's Lua takes only a few dozen children (each costs one of luabind's few dependency slots; past them bpp stops with an assertion in `object_rep.cpp`). Build big bodies as trees of small compounds, neighbours together (Lehmer's sieve's `tree()`), and don't add and remove children over and over: move them (`updateChildTransform`), as the sieve's pegs do.
- **Cylinders:** bpp's Lua has `btCylinderShapeX` and `btCylinderShapeZ` but no constructor for the upright `btCylinderShape`: stand a Z one up.
- **Convex pieces** from a 2D outline: `exhibits/LehmerSieve/parts/make-geometry.py` (shapely's constrained triangulation, merged while convex); check that the pieces' areas add up to the outline's. Corners inside an outline cost pieces; draw holes and inner arcs coarser for the contacts than for the print.
- **An OpenSCAD part made a body** must be set to collide before `v:add` (the exhibits' `part()` turns collisions off).
- `v:eachContact(fn)` calls `fn(a, b, x, y, z, nx, ny, nz, distance, impulse)` for every contact: the quickest way to find what jams.
- **Geneva drives** index well but their locking disc only bounds a star's turn (about a degree either way): a sprung detent per star puts it on its place, and the disc's relief must clear the star's lead and lag (`exhibits/LehmerSieve/FINDINGS.md`).
- `.gitignore` at the top ignores `*.scad`: the hall's own `.gitignore` has `!*.scad`, and STL files are not kept (each `make-stls.sh` makes them).

6. How to make an exhibit

1. Pick it from `MACHINES.md`; it goes in the bay its interest says (the walk rises), which may move others along (`hall.lua` EXHIBITS; README table; PDF subtitles in `make-pdfs.sh`).
2. Prototype the mechanism in a scratch script first and measure it.

3. `exhibits/<Name>/`: the script runs on its own and in the hall (find its folder with the `DIR` loop the others use; `hallPath` for OpenSCAD's full path), uses `exhibit-table.lua` (the machine's footprint), the start hold, resting, a tour that prints `RESULT ... PASS/FAIL`, sliders, keys, help text and hover text.
4. `parts/*.scad` as the single source of geometry (or, as the Pascaline, a Lua geometry file that writes the SCAD); `make-stls.sh` with each part turned to print; check the STLs and overlaps.
5. Sweep friction and speed, record it in `FINDINGS.md`; `README.md` with the parts table, what to buy, assembly; both PDFs; a `make-pdfs.sh` line.
6. In the hall: its EXHIBITS entry (`floorY = -100`), placard lettering (`make-labels.py`), `tools/check-all.sh` (add it there), screenshots.
7. **Make it show its working in the 3D view from the start**, not only in the console and tooltips: William watches in the GUI. Lehmer's sieve first showed nothing of N, x or the answer, and its probe's depth was hidden; it got a board of number drums (the exhibit's, said so) and a printed depth scale. When a machine runs on by itself, pause between steps so each reading can be seen (the sieve's "through" drum was cleared in the frame it was set). Say what the machine computes and what the script does for the operator (the sieve only says "maybe"; the square test is Lua's).

7. Where things stand and what's next

- **Done (2026-10-21): Lehmer's number sieve** (east 1): four Geneva stars of 9, 8, 7 and 5 places on one axle, driven from one shaft, detents to centre them, pegged holes, a cam-lifted probe; it factors 1241 (and 1679, 1157, 2813: `N`) by Fermat's way. **2026-10-25:** William found it hard to see what it computed: now a board beside it (the exhibit's) shows N, x, each wheel's hole under the probe (green open, red pegged), the depth, and at a stop x*x - N, the verdict and the factors; a printed depth scale on the probe's guide reads how far the probe went; `T` runs the tour for the number `N` chose. The wheels' colours and the see-through cam are still first guesses. 2026-10-26:* the depth now stays on the board between turns, and the camera is further back; William's window is narrower than the virtual screen (about 1450 px of 3D view, not 1658), so frame views with room to spare. He asked whether the cam is one print: it is (one watertight solid); only the simulation's contact shape is 72 convex wedges. Offered, not done: finer cam steps (5 degree flats now). It is the dearest exhibit to step (a 25 s tour took about three minutes here, headless): if it is slow on his computer, `LS_HZ=1200` halves that and still passes.
- **Done (2026-10-26): the Digi-Comp II, step 1, the counter** (west 3): three printed rockers, tipped over by the balls alone, and a feeder wheel. **The same day, step 2a, the accumulator,** which replaces it. William asked for the rest of the real machine; its manual (Evil Mad Scientist's copy, cdn.evilmadscientist.com/KitInstrux/DCII-manual.pdf) gives the registers: accumulator 7 flip-flops, memory 4 bits, MQ 3, distributor 3 (a ring counter), mode flip-flops T1 to T6 and complement CF1, CF2; switches MULTIPLY, CLEAR, COUNT, COMPLEMENT, OVERFLOW (run/halt) and AUTO/MANUAL; the ball's path INPUT, MULTIPLY, CLEAR, COUNT, then down the accumulator. Built now: a **two-level** board (balls that are done drop through holes to the lower level and roll to the collector), the 7-bit accumulator (each rocker has a foot on the lower level for CLEAR: a wedge a clear ball pushes back to 0), hand-set **switches** (flaps) for CLEAR and COUNT, and START (the feeder wheel, turned by hand). Keys: Space START, C and N the switches, 1 to 7 set a bit by hand. Board 322 x 878 mm, 10 upper and 10 lower tiles; table 100 x 150.
- **Done (2026-10-26, while William slept): step 2b, add and multiply.** The memory (four hand-set switches beside A1 to A4: 1 sends the ball into that flip-flop's funnel, by a long left wall the carry doesn't cross; 0 drops it), the MQ (three flip-flops read the other way, so a ball takes one away; a borrow out of it drops through STOP and the hand stops), the distributor (a tree of three flip-flops: the MQ and M4, M1, M3, M2 by turns; the manual's is a ring) and MULTIPLY, all thirteen flip-flops the

same part. CLEAR's "on" outlet is on its right now (the memory's lane needs A1's left). Tour: count, clear, 37 + 11 = 48, 13 x 3 = 39, every ball checked; the hand puts balls back from the collector. Keys: M, C, N; 1 to 7; H J K L the memory; 8 9 0 the MQ; D the distributor. Board 424 x 1216 mm, 14 + 14 tiles; table 110 x 180. Not yet seen by William.

- **2026-10-27:** William asked for an illustrated manual of every operation, now and to come, and which parts each needs (for building part of the machine): `exhibits/DigiComp2/MANUAL.md`. Its figures are drawn from the geometry. Keep it up to date as steps are added; its section 6 (complement, subtract, divide, AUTO) gives E.S.R.'s ways and this board's.
- **Done the same night: the narrated tour.** Before each ball a yellow dashed line on the board shows the way it will go, and the sign's THIS BALL says what it will do and what A was and will be (`route()`, `preview()` in `digicomp2.lua`; also on `Space`). `DC_STEP` starts the tour at a later step, for testing.
- **Done (2026-10-27): step 2c, the complement: subtract and divide.** William chose the longer board (more visible) over a lower-level inverting lane. Not E.S.R.'s T1 to T6, CF1, CF2: under each of A1 to A6's done holes a **done lane** runs on into the next flip-flop's funnel, and six **shutters** (one bar, high over the walls: the COMPLEMENT switch, `X`) cover the holes, so a ball goes on through every flip-flop to A7 and turns each over. Complement = two balls, the hand moving COMPLEMENT between them (CF1 and CF2's job). Dividing: MQ 7, the hand watches OVERFLOW and stops at the end of that cycle; quotient and remainder from the cycles and A (MANUAL section 6; FINDINGS says why the end of the cycle). A's steps are 14.75 cm apart now (`G.ADZ`), all its funnels have both walls long. Tour adds the complement of 25, 12 - 4 and 13 / 5 (about 27 600 frames; check-all allows 29 000). Board 424 x 1573 mm, 18 + 18 tiles; table 110 x 220. `tests/complement.lua`. The camera stands farther back for the longer board.
- **Done (2026-10-28): the hall loads a bit at a time.** William found the hall slow to start. It is shown at once now, a sheet over each exhibit's bay, and the exhibits load one a frame (the one you're at first); the status bar says which. The time is OpenSCAD making parts on a cold cache (94 s for all five here; 1.8 s cached). The Digi-Comp's sign letters are in `parts/sign.scad` now, so changing its geometry no longer remakes them.
- **Done the same night: programs.** William saw that the procedures are the operator's part (the machine has no stored program: its switches are control lines, each ball a clock tick, the operator its control unit) and asked for them as scripts the exhibit reads and runs line by line, with a library. `programs/*.dc2` (listed in `programs/LIBRARY`): the keys, plus `SET`, `UNTIL STOP|OVERFLOW|CYCLE`, `CHECK`; the "program" slider and `G` run one, the listing in the Shortcuts pane and the line in the status bar; `DC_PROGRAM=all` runs the library (53 000 frames, 12 minutes here). MANUAL section 9. (The hall's and the exhibits' status lines were never shown: bpp's `statusText` is a property, not a `setStatusText` method; the hall passes it on now.)
- **Then: the program board.** A board on a stand behind the machine shows the program, its line lit, scrolling with it; `B` flies the camera to it and back (set the camera's up vector, then its place, then its look, as separate statements: Lua's multiple assignment has no order).
- **Then: the tour is the library, the meter is off.** William asked for the tour to run the programs (so what starts by itself is a program, line by line), and for the hall's cost meter to be off unless asked for (the "Hall: meter" slider at the views, or `HALL_METER=1`; the freeze log goes with it). The Digi-Comp's tour is now ~53 000 frames; check-all allows 56 000 (and no longer runs the library separately).
- **Then: AUTO (step 2d).** The escapement is on the board (the knob is gone: the weight, a torque, turns the wheel, the fly caps its speed, the anchor holds it; Space pushes the START lever). Every ball goes down a trip channel at the bottom right, under a trip lever (an arm trailing down the slope from an axle: a ball can only lift it; its underside stays high near the axle, so a ball meets it far out), to the store. The axle comes out through the rim; an arm hangs from it beside the board, and a thread from its foot runs up beside the board to an eye and across to the START lever, which it pulls as the hand

does. AUTO (`U`): the thread hooked in; MANUAL: off. STOP's ball drops into a pocket and trips nothing. Programs have `AUTO`, `MANUAL` and `RUN` (each ball checked at the lever, the next in flight); `multiply-auto.dc2` does 13 x 5 in 21 balls from one START. The library (the tour) is ~81 000 frames; check-all allows 86 000. FINDINGS, step 2d, has the tries: a lever reaching up the slope jammed; a push rod up the board worked but was too heavy to push unless balanced, and William pointed out a thin rod 1.5 m long can pull but hardly push, hence the thread. In the simulation the thread is inextensible and pull-only (impulses each step, `trip.pull`). The weight and fly as parts are not done.

- **Then: OVERFLOW's halt (step 2e).** OVERFLOW is a shutter over the overflow hole (key `O`, program step `OVERFLOW HALT|RUN`): RUN, the carry out of A7 rolls over it to the collector; HALT, it drops into a pocket and trips nothing, so AUTO stops. `count-out-auto.dc2` and `divide-auto.dc2` run E.S.R.'s way. A halt mid-cycle can leave A = 0 when a division isn't exact (11 / 5): the manual says to finish the cycle by hand. The tour is ~88 000 frames; check-all allows 96 000 frames and two hours a run. **Not yet checked whole:** the two new programs passed on their own, and check-all's four other exhibits passed, but its Digi-Comp run (the library) went past the old hour's timeout here with the machine loaded and was stopped. First thing in a new session: `./tools/check-all.sh` and `tools/check-lua52.sh` from BabbagesHall, and fix whatever fails.
- **Next: the weight and the fly as parts (step 2f).** William chose a bought rotary damper for the fly: Fuji Latex (ACE) **FRT-G2-200**, 0.2 N cm at 20 rpm, on the wheel's shaft 1:1; the weight a cup of nine balls (~76 g) on a cord from a 4 mm drum under the plate, over a pulley at the top edge, with a ratchet for winding. The specs, the sums and what is open are in FINDINGS, step 2e, "Planned for the next step". In the simulation: drag c w (c = 0.96 mN m s) for `FLY_W`'s cap; run the library at c +- 35%. A possible tighter rule for dividing in AUTO (A = 0 at a halt is exact only if M's bits still to come in that cycle are all 0) was offered to William, not yet taken.
- **Then** each program showing only the parts it needs (William: "just add the parts you need when you need them", on the full board, not a compacted one). William pointed to a replica: www.instructables.com/Digi-Comp-II-Replica/ (only its summary could be fetched here).
- The other 39 machines in MACHINES.md have no bays: adding any grows the hall by a row of two, each going in by its rank (the Galton board, 13, and the planimeter, 14, would go in after the sieve, before the Pascaline).
- **Then, in the order of the covered bays,** each its own step: **Kelvin's tide predictor** (east 3: cranks and a bead rope over pulleys), **the Differential Analyser** (west 4) and **the Difference Engine** (east 4). New bays (the hall grows by a row of two) would suit the Galton board and planimeter (quick and clear) or the Enigma and the Antikythera mechanism (striking, within reach).
- **Settled (2026-10-21):** William has tried the Pascaline (the mouse and the keys) and it works well; the stylus collision he thought he saw wasn't one. Euler's Gallery keeps its own tables as they are; the exhibit standard is the hall's. The Rec Room's camera flights look good, the top-down table view included.
- **Open, for later:** POV-Ray finds textures only with `+L` the bpp folder (their paths are relative to it): Jakob has it. A printed detent (`detent.stl`, 1.2 mm leaf) would test the stiffness worked out from PLA; William will print one some other time.

8. Opening message for the next chat

> Clone github.com/bullet-physics-playground/bpp and read demo/WyomingWill/BabbagesHall/HANDOFF.md, then MACHINES.md and README.md there. Build bpp as described, run tools/check-all.sh, and continue with HANDOFF.md section 7: [the next step, e.g. the Digi-Comp II's AUTO]. Deliver each step as a patch, with its documents and PDFs, and the git commands.